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MILITARY SPECIFICATION

ROTATING BANDS, WELDED OVERLAY, FERROUS AND NONFERROUS, FOR PROJECTILES

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers ferrous and nonferrous welded overlay rotating bands for projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-I-6866 — Inspection; Fluorescent Penetrant, Method of.

MIL-M-11472 — Magnetic-Particle Inspection; Process for Ferro-Magnetic Materials.

STANDARDS

FEDERAL

FED TEST METHOD STD. No. 151 — Metals; Test Methods.

MILITARY

MIL-STD-20 — Welding Terms and Definitions.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Prior to fabrication on any contract or order, the contractor or manufacturer shall submit sample welded overlay banded projectiles for inspection, through the procuring agency, to the designated Government facility. The sample shall be chosen and disposed of in accordance with table I. Lot size may be presumed. The sample shall be manufactured in the same manner as regular production lots.

3.2 Process. The overlays produced under this specification shall be produced by a welding process approved by the procuring agency.

3.3 Materials. The welded overlay rotating band shall be fabricated from soft iron, copper, or other nonferrous material as authorized by the procuring agency. The projectile to which the overlay is applied shall be of material, type, and design as specified by the procuring agency.

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(Number of projectiles to be used for inspection under various paragraphs.)

TABLE I. *Disposition of samples*

Lot size	Sample size	4.5.2 & 4.5.4	4.5.3	4.5.5	4.5.6	4.5.7
25 to 40	7	7	3	2	2	[1]
41 to 65	10	10	3	3	3	[1]
66 to 110	15	15	3	3	3	[1]
111 to 180	23	23	4	4	4	[1]
181 to 300	35	35	5	5	5	[1]
301 to 800	50	50	6	6	6	[1]
801 to 3,200	75	75	8	8	8	[1]
3,200 to 8,000	100	100	10	10	10	[1]

[1] The same projectiles used for inspection under paragraphs 4.5.4, 4.5.5, and 4.5.6 may also be used for inspection under 4.5.7 since their value for tensile tests will not be altered by the preceding tests.

3.4 Projectile properties. The mechanical properties of the finished projectile in the area beneath the rotating band shall meet the requirements of the applicable projectile specification for that region.

3.5 Depth of fusion.

3.5.1 *Ferrous overlays.* The depth of fusion of ferrous overlays shall not be greater than $\frac{1}{16}$ inch below the original surface of the projectile.

3.5.2 *Nonferrous overlays.* The depth of fusion of nonferrous overlays on projectiles with wall thicknesses of $\frac{1}{8}$ inch or less beneath the band shall not exceed $\frac{1}{64}$ inch. In projectiles with wall thicknesses greater than $\frac{1}{8}$ inch but less than $\frac{1}{4}$ inch beneath the band, depth of fusion of nonferrous overlays shall not exceed $\frac{1}{32}$ inch. In projectiles with wall thicknesses of $\frac{1}{4}$ inch or greater beneath the band or in solid projectiles, depth of fusion of nonferrous overlays shall not exceed $\frac{1}{16}$ inch. (The limits set for depth of fusion in 3.5.1 and 3.5.2 shall be adhered to unless otherwise specified by the procuring agency).

3.6 Hardness of the finish machined rotating bands.

3.6.1 *Ferrous rotating bands.* The surface of the finished ferrous rotating band shall not exceed a hardness of Rockwell B70 when tested as specified in 4.5.4.

3.6.2 *Nonferrous rotating band.* For copper rotating bands or rotating band sections (as separated by cannelures) which are less than .3 of an inch in width at the surface being tested or less than 0.040 of an inch in thickness, the Rockwell 30T scale shall be used and the hardness shall not exceed Rockwell 30T36 when tested as specified in 4.5.4. Copper rotating bands or rotating band sections which do not require the use of Rockwell 30T scale as specified above shall be tested with the Rockwell E scale and the hardness shall not exceed Rockwell E75 when tested as specified in 4.5.4. The surface hardness of other materials finished as rotating bands shall not exceed the specific limit authorized by the procuring agency.

3.6.3 *Thermal treatment.* A thermal treatment of the weld deposit is permitted to meet the required hardness provided the thermal treatment does not reduce the properties of the projectile below those specified on applicable drawings. However, the process used shall be subject to the approval of the procuring agency.

3.7 Alternate chemical requirements. In the event that a test lot of nonferrous welded overlay rotating bands does not meet the hardness requirements of 3.6, the hardness requirements may be waived and chemical analysis of the bands shall be used to deter-

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mine the acceptability of the lot. The iron content of nonferrous bands shall not exceed 1.25 percent when the chemical analysis is carried out in accordance with 4.5.5. This limit for iron content may be raised at the discretion of the procuring agency.

3.8 Voids at band surfaces. The overlay deposit, when machined to the final specified dimensions, shall not reveal on the surface any cavities that exceed 1/16 inch in greatest dimension. The size and number of cavities equal to or less than 1/16 inch in greatest dimension within 1 square inch of surface area excluding cannelures, if present, and as measured by 0.019 and 0.039-inch diameter probes, shall not exceed:

(a) 1 cavity 0.040 inch in diameter or greater, or

(b) 4 cavities 0.020 inch in diameter or greater, or

(c) 8 cavities less than 0.020 inch in diameter, or

(d) a total of 10 cavities.

3.9 Voids at weld metal-base metal interfaces. The sum total of the greatest dimensions of all voids appearing at the weld metal-base metal interface, when observed as specified in 4.5.3, shall not exceed 25 percent of the continuous length of the interface at which they appear. (The continuous length of an interface for the purposes of this specification is defined as that observable in a transverse cross section of a finished rotating band having no defects or discontinuities other than those specified in the applicable drawing).

3.10 Slag. No slag shall be permitted in any overlay deposit when examined in accordance with 4.5.6.

3.11 Cracks. Cracks in the finished machined band shall not be permitted as observed visually when examined as specified in 4.5.2. In case of dispute, the rotating band shall be submitted to magnetic-particle or fluorescent-penetrant inspection in accordance with specifications MIL–M–11472 and MIL–I–6866, respectively. There shall be no underbead cracking or cracks extending into the base metal (see 4.5.3).

3.12 Finish. The weld deposit shall be capable of being machined to the dimensions of the applicable drawings. When specified by the procuring agency, finish machined bands shall be chemically treated and painted in accordance with the requirements specified by that agency. Weld metal spatter on projectiles shall be removed to within the dimensional tolerances of the applicable projectile drawings unless otherwise specified by the procuring agency.

3.13 Workmanship. The finished rotating band shall be clean and free of rust, scale, or other defects that may interfere with the performance or otherwise make the projectile hazardous for use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classes of inspection. All examination and testing shall be to determine conformance to the requirements of this specification to serve as a basis for acceptance.

4.3 Lot for purpose of welded band inspection. A lot for use in inspection of welded overlay band quality shall consist of not more than the number of projectiles of the same size, shape, and design which are overlayed in a 40-hour work period on a single overlay banding machine with no change in process or raw materials. Smaller quantities may con-

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stitute sublots at the discretion of the inspector. Projectiles overlayed on separate machines shall constitute separate lots.

4.4 **Sampling.** After final machining of the weld deposit, samples shall be selected in accordance with table I for examination to determine conformance to section 3. All the samples shall be selected by the inspector so that the entire cross section of the lot shall be represented. Disposition of the sample shall be in accordance with table I. In the event that chemical analysis is required (see 3.7), the projectiles which showed the highest average band hardness shall be used to obtain samples for chemical analysis.

4.5 **Acceptance tests.**

4.5.1 *Preproduction sample.* The projectiles of the preproduction sample shall be tested in the same manner as a sample selected from a regular production lot and shall be inspected by the designated Government facility for conformance with section 3 by the methods outlined in section 4.

4.5.2 *Voids, overlay cracks, finish, and workmanship.* The projectiles selected as a sample under 4.4 shall be examined visually for voids at the band surface (see 3.8), cracks in the overlay (see 3.11), finish (see 3.12), and workmanship (see 3.13).

4.5.3 *Depth of fusion, base metal cracks and voids at the weld metal-base metal interface.* The projectiles selected for measurement of depth of fusion, base metal cracks and voids at the weld metal-base metal interface shall be sectioned and the band area suitably polished and visually examined for compliance with 3.5, 3.9, and 3.11. In projectiles with wall thickness of ¼ inch or less, the projectile shall be sectioned longitudinally through the band area into four approximately equal quarters. One of the two surfaces produced by each quartering cut shall be polished for examination. In projectiles with wall thicknesses greater than ¼ inch or in solid projectiles, the projectile shall be sectioned longitudinally through the band area so as to yield approximately equal halves. The two band areas on one of the halves shall then be polished for examination. The forward and rear portions of the projectile may be removed prior to sectioning. Cuts made for this purpose, however, shall be at least ½ inch from the nearest edge of the finished rotating band.

4.5.4 *Overlay band hardness tests.* The finish machined bands of the projectiles selected in 4.4 shall be tested for hardness in at least 6 locations, 60° apart by methods outlined in Fed. Test Method Std. No. 151. The average of the readings on any one band shall be considered the hardness of that band for purposes of compliance with 3.6. The Rockwell E scale shall be used for testing any copper rotating band which contains any section permitting its use. (See 3.6 for dimensional requirements of a copper rotating band section which permits the use of the Rockwell E scale.)

4.5.5 *Alternate chemical requirements.* From the projectiles selected under 4.4, samples shall be obtained from each band by machining the finished band concentrically with the projectile to a diameter of 0.020 of an inch greater than the projectile diameter beneath the band. All chips shall be collected. Each complete specimen thus obtained from each band shall be dissolved and the analysis made from the individual solutions. Chemical analysis shall be conducted in accordance with Fed. Test Method Std. No. 151 except that samples may not be cleaned with a magnet.

4.5.6 *Evidence of slag.* The rotating bands of projectiles selected for evidence of slag in the weld deposit shall be examined by being turned down in a lathe taking 0.010 inch depth of cut for each examination. The overlay shall be turned down to the maximum depth of fusion. The lot shall be considered acceptable provided the requirements of 3.10 are met.

4.5.7 *Projectile tensile tests.* The projectiles used for determining evidence of slag, hardness and, when necessary, chemical composition under 4.5.4, 4.5.5, and 4.5.6 shall also be

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used for projectile tensile tests. Tensile tests shall be conducted in accordance with applicable projectile specifications.

4.6 Acceptance. Welded overlays on projectiles in lot or sublot shall be considered acceptable provided the requirements outlined in section 3 are obtained.

4.7 Retests. Welded overlay bands rejected due to defects or other imperfections may be reworked and resubmitted for tests as an individual mixed lot. However, all projectiles contained in such a mixed lot shall have been rewelded the same number of times. Rejected lots may also be reworked and resubmitted. All tests specified for the original lot shall be applied and these tests shall show acceptable quality.

5. PREPARATION FOR DELIVERY

5.1 Not applicable.

6. NOTES

6.1 Intended use. The welded overlay band serves two functions. It is engaged by the rifling in the gun tube and thus imparts rotation to the projectile and it obturates the propellant gases. Both of these functions are highly necessary to the satisfactory performance of a projectile.

6.2 Difference between a contractor and a manufacturer. A contractor is an organization having a direct contract with a government agency, whereas the manufacturer is an organization having a contract with a government contractor. A contractor may also be the manufacturer.

6.3 Explanation of sections 3.5, 3.6, and 3.7. Those familiar with welded overlays rotating bands are aware that relationships exist between depth of fusion, band hardness, and band composition in both ferrous and nonferrous bands. Hypothetically, then, it should be possible to place limits on one of these variables and thus control the other two. Unfortunately, however, certain factors prohibit such simplification. The impossibility of accurately measuring the volume of base material incorporated into the overlay due to fusion negates the possibility of using depth of fusion as a sole criterion. Work hardening of the band due to machining operations and irregular dispersion of diluted base material influences band hardness. Thus, the relationship between hardness and the average depth of fusion and dilution may be inconsistent. Chemical analysis would accurately limit composition but any requirement set on this basis alone could not detect cases of severe localized base metal fusion. For these reasons, it has been necessary to incorporate all three factors into the specification.

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